

March 15, 2025

The Suzanne H. Plimpton

Networking and Information Technology Research and Development (NITRD) National
Coordination Office (NCO), National Science Foundation
2415 Eisenhower Avenue,
Alexandria, VA 22314, USA

Re: Comments in response to the Request for Request for Information on the
Development of an Artificial Intelligence (AI) Action Plan (Vol. 90, No.24
Federal Register, Day, Date, Request for Comments) Docket No.: FR Doc. 2025-
02305

Dear Officer Plimpton:

As a member of the public, I am offering my comments to the Office of Science and Technology Policy (“OSTP”) in response to the request for comments related to the Office’s inquiry on the development of the AI Action Plan.

As explained further in my below responses, I maintain several policy positions regarding the priorities of the AI Action Plan. First, the development of the AI Action Plan should address the energy consumption of power plants for AI Development and Use. Second, the development of the AI Action Plan should consider research and development in materials characterization for AI development. Third, the development of the AI Action Plan should adapt patent law and policy to AI to ensure incentives for the development and appropriate use of AI.

I. Addressing Energy Consumption of Power Plants for AI Development and Use

1. Introduction.

The growth of artificial intelligence (AI) technologies comes with surging energy demands, particularly in the infrastructure that supports AI’s development and use--data centers. These data centers, essential for powering AI development and applications, are placing significant pressure on the nation’s energy sources including fossil fuel-based sources. The electricity demand in data centers mainly stems from computing and cooling. First, AI’s development requires massive computational power, for example when dealing with large datasets or running complex AI models, such as those used in materials discovery. The high computational demands significantly increase energy consumption, as more powerful servers are used to process data. Second, cooling technologies are needed to manage the heat produced by the dense concentration of data center equipment. As data centers and their equipment grow, cooling demands intensify, often relying on water from already stressed watershed areas.

As AI models continue to grow larger and more complex, so too will their energy demands. The growth in energy consumption raises economic and pollutant concerns, both of which pose challenges to national energy security and independence. As such, addressing these energy demands without compromising energy stability, affordability, or the environment has become a pressing priority.

2. Importance of Addressing Energy Consumption for AI Development and Use

Economic considerations of rising energy costs: The increasing demand for electricity to power AI-related data centers is likely to lead to higher electricity prices, putting pressure on consumers and businesses. As AI infrastructure expands, fossil-fuel-powered plants, currently the primary energy source for much of this infrastructure, will need to burn more coal or natural gas to generate additional electricity. In response to the growing demand, power plants may need to upgrade or expand their infrastructure, leading to increased capital expenditures. These higher fuel consumption and capital expenditures contribute to rising electricity prices, which can directly impact consumers and industries that heavily involve AI in their operations. Businesses that use AI in their daily functions may experience increased operational costs, which can reduce their profit margins and global competitiveness.

Pollutant considerations: Fossil-fuel power plants to meet the energy demands of AI development emit harmful substances, such as sulfur dioxide, nitrogen oxides, and particulate matter, all of which are regulated under the Clean Air Act. To mitigate these conventional pollutions and carbon dioxide, power plants install emission-control technologies, such as carbon capture technologies, scrubbers, and desulfurization systems. However, these technologies often require a portion of the plant's energy output to operate that increases operational costs and reduce the overall energy efficiency. For instance, the energy dedicated to carbon capture processes, including energy-intensive steps like recharging the chemicals to separate carbon dioxide, cannot be sold to the grid. As a result, power plants may need to raise electricity prices to offset these added operational costs. This makes power plants challenging to adopt carbon capture technologies at scale.

Energy Security and Independence: The increasing reliance on fossil fuels for AI-related power consumption can heighten domestic energy dependence and vulnerability to global market fluctuations. To meet the rising energy demands of conventional and emerging technologies like AI, a reliable and affordable energy supply is essential. Therefore, this reliance needs to be managed efficiently.

3. Suggested Action for A Balanced Approach to Energy Efficiency for AI Infrastructure

Energy Efficiency Improvements: Support the development and implementation of pollution-control technologies that improve the energy efficiency of power plants supporting AI infrastructure. This includes integrating emissions control technologies like carbon capture and ensuring that cost-efficiency and operational viability are not compromised. A commonly used metric to assess the economic aspect of these technologies is energy use share, which measures the percentage of a power plant's energy output dedicated to the carbon capture process. A high energy use share could indicate low operational efficiency due to the additional energy required

for carbon capture and separation, impacting a plant's economic viability. However, a low energy use share does not necessarily equate to greater energy efficiency, as it overlooks the technological efficiency of the carbon capture process itself, in terms of the energy consumption per unit of carbon dioxide captured.

Balancing the operational efficiency of these technologies with their ability to effectively reduce emissions is key to making them both economically and environmentally viable. One element of this strategy is finding an optional balance point between operational efficiency and technological efficiency in emission control systems in terms of energy efficiency. For example, just as limiting the amount of power used in AI training and processing helps reduce energy consumption, power plants can limit emission control efforts to a balanced range. Rather than pushing for a nearly 100% emissions capture, which can be excessively energy-intensive and costly, a targeted, lower percentage of carbon dioxide absorption can be set that balances technological efficiency with operational costs. By taking into account both operational and technological efficiency, such approach can offer a more balanced pathway to improving energy efficiency in power plants supporting AI infrastructures without overburdening the system with emissions control technologies that unnecessarily increase costs for consumers.

Diversifying Energy Sources and technologies for power infrastructure: Encourage the integration of an energy mix that includes solar and geothermal, while also leveraging energy storage technologies to enhance power supply reliability and flexibility. This can include incorporating energy storage technologies such as lithium-ion batteries to ensure continuous uptime, which is crucial for AI workloads during power outages or disruptions. An example can be drawn from Google's use of scaled lithium-ion battery backup units in its data centers as a reliable power sourceⁱ. By combining different energy sources with energy storage technologies, AI infrastructure can better manage price volatility in fossil fuel markets and improve energy security and independence.

II. Addressing Research and Development in Materials Characterization for AI Development

1. Introduction

The AI action plan should recognize the key role of materials science, particularly materials characterization, in advancing AI technologies. Materials characterization techniques provide essential insights into the structural, properties of materials, which are fundamental for innovating and optimizing AI hardware, as well as enabling AI systems to reach their full potential.

First, AI needs Advanced Materials for Progress. AI systems, particularly in areas like quantum computing, depend on innovations in material properties to enhance performance. For example, in quantum computing, the performance of qubits is deeply linked to the properties of the materials used. Techniques like scanning probe microscopy are essential for understanding materials at the atomic and subatomic levels. These methods allow for the detailed examination of materials, enabling the design of hardware—such as chips, sensors, and memory devices—that supports next-generation AI technologies. Without these insights, the hardware powering AI will not be optimized, ultimately limiting AI's capabilities.

Second, AI's power is greatly amplified when coupled with the ability to predict and optimize materials' behavior at scale. Microscopy techniques like SPM can offer the high-resolution, data-rich insights required to train machine learning models. These models, in turn, predict the behavior of new materials including those that are better suited for AI applications. Therefore, fostering R&D in materials characterization is key to ensuring AI models have the comprehensive data needed to make more accurate predictions and drive advancements in materials discovery and design.

2. Importance of Addressing Research and Development in the Area of Materials Characterization

National Security: Materials characterization is vital for enhancing U.S. national security, particularly as AI technologies and advanced materials intersect with energy systems. For instance, microscopy techniques like scanning probe microscopies reveal structural and electronic properties, as well as the electric field information of perovskite-based photovoltaic systems. Surface and interface characterization tools help understand the interfaces between perovskite materials and traditional electronic materials. This understanding of material properties at the atomic and device levels forms a quantitative basis for the development of predictive models like AI to design materials with targeted properties, such as those suitable for energy applications. By strengthening materials characterization, the U.S. developing advanced materials critical to meet the U.S.'s national goals.

National Competitiveness and Technological Leadership: Strengthening the research and development in materials characterization helps ensure U.S. leadership in AI and technological innovation. As the performance of AI systems heavily relies on hardware such as processors, memory, and sensors, understanding the atomic and device levels of materials becomes essential for optimizing performance and driving technological advancements. This understanding enables the design of advanced materials for AI hardware and facilitates the development of predictive models powered by AI that can guide the design of materials with specific, targeted properties. Maintaining leadership in materials characterization ensures that the U.S. remains at the forefront of AI development and global technological competitiveness.

3. Suggested Action for Strengthening Research and Development in Materials Characterization

To ensure that the U.S. remains a global leader in the development of AI and AI-driven materials discovery and design, the AI action plan needs to support research networks, including the National Labs and universities. These research networks play a key role in advancing the nation's foundational technological capabilities, including materials characterization, necessary for AI innovation. This can include the design, development, and deployment of new materials characterization techniques and surface characterization techniques. This can also include reviewing the adequacy of current research infrastructure at National Labs and universities and expanding such capabilities to ensure sufficient foundational technological capabilities to solidify U.S. leadership in AI development and use.

III. Adapting Patent Law and Policy to AI to Ensure Incentives for the Development and Appropriate Use of AI

1. Introduction

As technology advances, especially with AI, it is essential to ensure that our patent system remains strong, and adaptable, and continues to incentivize innovation. The cost and complexity of developing AI are rising, making it critical to protect inventors' intellectual property. Patents have long been the engine driving innovation that benefits society. But as AI becomes more involved in the invention process, questions like who owns the rights to AI-assisted or AI-generated inventions could introduce uncertainty in the patent system. If not timely addressed, this uncertainty could undermine confidence in the patent system and slow progress in both AI development and its appropriate use in innovation.

2. Importance of Addressing Patent Law and Policy to Balance AI Development and Use

Impact on AI Development: Current inventorship criteria, which require a significant human contribution to claim patent rights, have been effective in motivating AI developers to advance AI technologies. Today, AI systems still rely on human developers to design, train, and guide AI models to perform tasks. For example, AI is used to predict material properties for materials discovery, a process that requires training AI systems on experimental and computational data of intrinsic materials information. This human involvement is crucial, as it enables AI to make predictions about the functional properties of new materials, thereby providing AI developers with the prospect of obtaining patents for their advancements.

Impact on the Ethical Use of AI in Innovation: AI's evolving capability, particularly in inventive processes, challenges traditional ideas of inventorship and ethical innovation. Currently, AI systems are tools that assist human inventors in the inventive process. But as they become more autonomous determining who should be credited as the inventor becomes complex. For example, when AI advances to the extent of autonomously running experiments in robotic laboratories for materials design, future AI users may find their contribution too insignificant to qualify for inventorship. Without clear guideline that takes into account to the pace of AI technologies, there is a risk that human inventors could be denied rightful recognition for AI-involved innovations, discouraging the use of AI in scientific progress. Additionally, the lack of clear guidelines around AI-driven discoveries could lead to manipulative practices, where human researchers may inflate their role in AI-driven discoveries to meet inventorship requirements. Such practices undermine the true potential of AI and can stifle genuine innovation.

3. Suggested Action for Addressing Patent Law and Policy to Balance AI Development and Use

Strengthen and Modernize the Patent System: To maintain U.S. leadership in global competitiveness, it is essential to ensure that patent law and policy evolve to reflect the growing capabilities of AI and its role in the invention process. This can include regularly assessing AI's capabilities and its impact on innovation and adapting the patent system accordingly.

Promote Ethical Practices in AI Innovation: Encourage the development of ethical standards for AI use in innovation. This can include ensuring transparency in how AI is employed in the inventive process and preventing the manipulation of inventorship by inflating human contributions in AI-driven innovations. Additionally, enhancing public understanding of the ethical implications of AI in scientific discovery can help ensure that AI contributes to innovation in ways that align with societal values and ethical standards.

IV. Conclusion

The author gratefully acknowledges the efforts of the Office regarding this comprehensive inquiry on artificial intelligence development plan. I thank you for the opportunity to provide such comments.

Statement:

This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.

Sincerely,

Sheng Tong

ⁱ Christina Peabody, Grogory Sizikov, “How we got to 100 million cells in our global Li-ion rack battery fleet,” February 25, 2025, <https://cloud.google.com/blog/topics/systems/100-million-li-ion-cells-in-google-data-centers>, (accessed March 15, 2025).